BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 141 · Ethical, Cultural and Environmental Impacts

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What does a compression utility do?

Q2. What does a denial-of-service attack try to do?
A. Make a service unavailable
B. Improve network speed
C. Create secure backups
D. Repair damaged hardware

Q3. “Data interception is only copying a paper document from a filing cabinet.” Correct this. Explain why.

Q4. A new user needs to open and sort files. “A user interface replaces all CPU instructions.” Correct this. How does an interface help?

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

Ethical impacts — Consider fairness, privacy and responsible uses of collected data.

Cultural impacts — Digital technology changes communication, access to information and how groups participate.

Environmental impacts — Device manufacture, energy usage and electronic waste all have costs.

Stakeholders — A balanced evaluation considers benefits, harms and who experiences each effect.

New learning

Ethical impacts: Consider fairness, privacy and responsible uses of collected data.

Cultural impacts: Digital technology changes communication, access to information and how groups participate.

Environmental impacts: Device manufacture, energy usage and electronic waste all have costs.

Stakeholders: A balanced evaluation considers benefits, harms and who experiences each effect.

Worked example

Evaluate a school app that tracks pupil location all day. Give a balanced response.

1. Benefit: can support safety or locating missing equipment if appropriate.

2. Risk: constant tracking may intrude on privacy.

3. Identify need for proportionality, consent/legal basis and secure access.

4. Give a justified judgement about limited, transparent use.

We Do · Practise with guidance

Guided 1

Discuss replacing paper letters with a school app.

Worked method: Provide a balanced conclusion.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed:

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BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Write the missing ELSE output for a user without internet access.

```
hasInternet = false
if hasInternet then
    print("Use website")
else
    ...
endif
```

write

Question 1

Explain one environmental disadvantage of frequent device replacement. [3 marks]

Question 2

Give an ethical concern about large-scale personal-data collection. [3 marks]

Question 3

Explain one benefit and one disadvantage of online-only school communication. [4 marks]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Exit Ticket · Answer before checking

A balanced evaluation considers benefits and harms for stakeholders.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. It reduces the storage space required for data, using a suitable lossless or lossy method.

Q2. A — Make a service unavailable

Q3. Capturing data during transmission, potentially exposing confidential information.

Q4. A user interface lets people use the computer, through GUI controls or typed commands. Other OS tasks manage devices, accounts, memory and files.

I Do / We Do pages · Guided practice

Q1. Provide a balanced conclusion.

You Do pages · Code First

```
hasInternet = false
if hasInternet then
    print("Use website")
else
    print("Offer telephone access")
endif
```

Expected result: Offer telephone access

You Do pages · Independent questions

Q1. Creates electronic waste and uses energy/resources in manufacture.

Q2. Privacy, informed use/consent or biased profiling, with contextual explanation.

Q3. Faster cheaper access for connected households; excludes families without dependable devices/internet.

Exit Ticket · Model answer

True

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.